

Reference excerpt 01-023

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can be described using the Womersley equation that is a linear differential equation where the unknown is the function w and the pressure gradient $(dp(t)/dz)$ is the source term: $2w(r, z, t) r^2 + 1/r \cdot w(r, z, t) r - w(r, z, t) t = 1/p(t, z) z$ (1) The radial component of the blood velocity and the non-linear terms of the equation are negligible for $c \gg w$.

[Womersley(1955c)] The solution of Eq.1 and the complete procedure to calculate it are explained in detail in [Sisini et al.(2015b)]. However the solution is there defined up to the multiplicative constant c and up to the additive constant w_0 . As a consequence, the instantaneous velocity trace, calculated in this way, is proportional but not equal to the actual blood velocity. Pressure waves propagation The pressure is transmitted along the direction AR-IJV according to the following equation $p z = + 1/c \cdot p t$ (2) For each cardiac cycle, the pressure in the RA reaches its maximum value between t_P and t_Q , where t_P indicates the ECG P wave and t_Q indicates the wave Q. In this note, for the sake of simplicity, it is assumed that the pressure in the RA is maximum in correspondence of the instant t_P Q which is the average between t_P and t_Q . The relationship between the pressure in the RA and the pressure in the point G at the same instant of time is given by: $p(t + lG/c, zG) = p(t, zRA)$ (3) Where